DSMM — Deployment State Monitoring Module

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OOF® Origin Open Foundation®

Independent Methodological Authority

Architecture: SLGA™ — Software Lifecycle Governance Architecture

Parent Standard: SDPS — Software Deployment Standard

Category: Governance & Enforcement

Subcategory: Software Deployment Governance

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Canonical Definition

DSMM — Deployment State Monitoring Module is the Core Module within SDPS that governs how the actual state of a software deployment is observed, evaluated, classified, and reassessed against its authorized scope, intended operational conditions, production version identity, verification requirements, and applicable intervention thresholds.

DSMM establishes deployment-state monitoring as an evidence-supported

governance mechanism rather than a collection of technical metrics, pipeline completion signals, or automated health reports.

It governs the relationship between deployment execution, production

version state, monitoring coverage, operational observations, expected conditions, material deviations, and the decisions required when the resulting operational reality cannot be sufficiently established or accepted.

The module distinguishes technical execution completion from verified

deployment establishment, monitoring activity from sufficient observability, software availability from operational acceptability, and observed anomalies from confirmed causes.

DSMM applies to manually operated deployments, continuous delivery systems,

distributed applications, cloud infrastructure, autonomous AI agents, embedded software, industrial systems, and robotic platforms.

Its principal governance output is the Software Deployment State Profile.

Operational Role

DSMM establishes and maintains an evidence-supported representation of the

actual deployment state.

It governs:

• Deployment-state identity

• Monitoring purpose and scope

• Intended operational state references

• Deployment execution observations

• Production version references

• Monitoring coverage

• Observation source identification

• Evidence reliability and freshness

• Deployment progression monitoring

• Operational condition verification

• Deployment-state consistency

• Monitoring gaps

• Deployment anomalies

• State uncertainty and conflicts

• Intervention thresholds

• Monitoring-driven escalation

• Deployment-state classification

• Material-change reassessment

• Historical state continuity

• Handoff to rollback and recovery governance

DSMM does not independently authorize deployments, redefine production artifact identity, execute rollback actions, or determine recovery success without the applicable evidence and governance references.

Operational Question

How can the actual state of a software deployment be observed, evaluated, and reassessed when execution outcomes, production versions, operational conditions, monitoring coverage, and material changes may differ from the intended deployment state, while ensuring that insufficient evidence, conflicting observations, unacceptable conditions, and intervention requirements remain identifiable and governable?

Position in Architecture Governance Flow

SLGA™ Governance Flow

Software Requirements → Software Creation → Software Dependencies →

Software Configuration → Software Quality → Software Security → Software Release → Software Deployment → Software Change Management → Software Retirement

Cross-cutting: Software Lifecycle Control

SDPS Core Module Flow

DPM — Deployment Planning Module → DAM — Deployment Authorization Module →

PVCM — Production Version Control Module → DSMM — Deployment State Monitoring Module → RRVM — Rollback & Recovery Module

DSMM is the fourth Core Module within SDPS.

It receives the intended deployment conditions from DPM, authorization

conditions from DAM, and production version evidence from PVCM.

It evaluates whether the observed deployment state supports continuation,

operational acceptance, restriction, intervention, rollback consideration, or recovery escalation.

Its outputs provide the operational evidence basis for RRVM and other

applicable software lifecycle governance mechanisms.

• Governed Space

Software Deployment State Monitoring Governance

This governed space concerns the observation, assessment, classification,

and reassessment of software deployment states within defined operational environments.

It begins when deployment execution or an applicable deployment-state observation becomes relevant.

It continues through deployment progression, activation, operational verification, exception handling, stabilization, and material deployment-related changes.

Monitoring may continue after technical execution is complete where the deployment's operational conditions, version consistency, or verification obligations remain unresolved.

The scope of DSMM is deployment-state governance, not unrestricted monitoring of every aspect of ongoing software operations.

Governed Object

The primary Governed Object is:

A Software Deployment State representing the evidence-supported

relationship between an identifiable deployment, its intended and authorized conditions, the software version actually present or operating, observed execution and operational conditions, monitoring coverage, and the current governance classification of the resulting deployment state.

The governed object must preserve:

Deployment State ID

Deployment ID

Deployment Plan Reference

Deployment Authorization Reference

Release ID

Environment ID

Production Version Reference

Monitoring Scope

Observation Time

Observed Conditions

Evidence References

Monitoring Coverage

Material Deviations

Intervention Status

Deployment State Classification

Version

Scope

DSMM governs:

Deployment State Identification

Monitoring Scope Definition

Observation Requirements

Execution State Observation

Production Version Association

Operational Condition Monitoring

Monitoring Coverage Assessment

Evidence Sufficiency

State Consistency Assessment

Anomaly Identification

Monitoring Uncertainty

Intervention Trigger Evaluation

State Classification

Monitoring Continuity

Reassessment

Historical State Preservation

Rollback and Recovery Handoff

DSMM does not prescribe a particular monitoring platform, observability framework, telemetry protocol, or infrastructure technology.

Foundational Principle

A deployment state must be established from evidence of the operational reality that exists, not merely from reports that deployment actions were completed or that monitoring systems are active.

Deployment Executed ≠ Deployment Successfully Established.

Monitoring Active ≠ Deployment State Fully Observable.

Service Available ≠ Deployment Operationally Acceptable.

No Alert ≠ No Material Problem.

Observed Condition ≠ Verified Cause.

Deployment State Identity

Every governed deployment state must have an identifiable reference.

Relevant elements include:

Deployment State ID

Deployment ID

Environment ID

Operational Segment Reference

Observation Scope

Observation Time

State Classification

Evidence Reference

State Version

State identity must remain distinguishable from deployment execution records and production version identities.

Monitoring Purpose

Monitoring purpose identifies the operational questions that deployment-state observations must answer.

These may include:

Did deployment execution reach the intended stage?

Is the expected software version present?

Is the intended version activated?

Are required services operating?

Are deployment conditions satisfied?

Are material deviations present?

Is the observed state sufficiently verified?

Is intervention required?

The monitoring purpose must reflect the actual deployment scope and consequences.

Monitoring Scope

Monitoring scope defines the environments, services, instances, devices,

and conditions subject to deployment-state observation.

Relevant elements include:

Environment

Operational Segments

Software Components

Production Versions

Relevant Dependencies

Configuration Conditions

Service Conditions

Deployment Stages

Verification Period

Material Restrictions

The scope must be sufficient to evaluate the claims being made about deployment state.

Intended State Reference

DSMM receives the intended operational state from DPM.

Relevant elements include:

Target Release

Target Production Version

Target Environment

Required Configuration

Required Dependencies

Required Activation State

Expected Service Conditions

Operational Acceptance Criteria

Verification Requirements

The intended state is a comparison reference.

It must not be treated as proof of the actual state.

Authorization Condition Reference

DSMM receives applicable deployment authorization conditions from DAM.

Relevant elements include:

Authorized Release

Authorized Environment

Authorized Scope

Authorized Execution Actor

Permitted Deployment Actions

Operational Restrictions

Validity Conditions

Mandatory Verification Conditions

Material Exceptions

DSMM evaluates observed deployment conditions against these references

without independently granting or revoking deployment authority.

Production Version Reference

DSMM consumes production version identities and consistency assessments

established through PVCM.

Relevant elements include:

Observed Artifact Identity

Installed Version

Activated Version

Runtime Version

Version Distribution

Artifact Verification Status

Version Consistency

Unknown Version Segments

Version Evidence

DSMM evaluates the operational consequences of the observed version state.

It does not replace PVCM's production artifact identity governance.

Deployment Execution Observation

DSMM must identify relevant execution events and their reported outcomes.

Examples include:

Deployment Started

Artifact Transferred

Installation Completed

Configuration Applied

Service Activated

Traffic Transitioned

Execution Paused

Execution Failed

Execution Completed

Execution events provide evidence about deployment progression.

They do not independently establish operational success.

Deployment Progression State

DSMM must represent material deployment progression states.

These may include:

Not Started

Preparing

Execution In Progress

Partially Executed

Execution Paused

Execution Failed

Execution Completed

Verification Pending

Operational Verification In Progress

Operationally Established

Intervention Required

Recovery In Progress

Progression states must not conceal uncertainty about the resulting operational condition.

Operational Observation

Operational observations provide evidence about the environment affected by

deployment.

Relevant observations may include:

Service Availability

Process Health

Application Errors

Request Success

Response Latency

Resource Utilization

Dependency Connectivity

Configuration State

Data Integrity Conditions

Security-Relevant Signals

Physical System Conditions Where Applicable

Observation requirements depend on the deployment's intended outcome and

operational consequences.

Observation Source Identity

Every material observation must remain associated with an identifiable source.

Relevant elements include:

Observation Source ID

Monitoring System

Observed Target

Observation Method

Observation Time

Collection Actor or Agent

Evidence Reference

Known Limitations

Source Reliability Reference

A monitoring source must not be treated as reliable solely because it is automated.

Monitoring Coverage

Monitoring coverage represents the portion of the relevant deployment state

that can be observed through available evidence.

Relevant dimensions include:

Environment Coverage

Service Coverage

Instance Coverage

Version Coverage

Configuration Coverage

Dependency Coverage

Operational Condition Coverage

Temporal Coverage

Recovery Condition Coverage

Monitoring coverage must be evaluated relative to the claim being assessed.

Coverage Sufficiency

DSMM must determine whether monitoring coverage is sufficient for the required deployment-state assessment.

Relevant questions include:

Are all critical operational segments represented?

Are required conditions observable?

Are observations sufficiently current?

Are relevant deployment stages covered?

Are known blind spots identified?

Are material failure modes detectable?

Can the evidence support the claimed operational state?

Monitoring Configured ≠ Monitoring Sufficient.

Monitoring Blind Spots

Monitoring blind spots are material portions of deployment reality not adequately represented by available observations.

Examples include:

Unobserved Service Instances

Disconnected Devices

Missing Regional Telemetry

Unavailable Application Logs

Unverified Configuration State

Unknown Production Version

Unmonitored Data Migration

Missing Physical-System Feedback

Blind spots must remain explicit when they affect deployment-state confidence or acceptance.

Observation Freshness

Deployment-state evidence must be evaluated for temporal relevance.

Relevant elements include:

Observation Time

Collection Delay

Processing Delay

Relevant State Change

Evidence Age

Expected Update Frequency

Reverification Requirement

A previously verified state may no longer represent current deployment reality after a material change.

Observation Reliability

DSMM must assess the reliability of material observation sources.

Relevant factors include:

Source Identity

Collection Method

Measurement Integrity

Known Failure Modes

Data Completeness

Clock Consistency

Source Availability

Independent Corroboration

Known Bias or Limitations

An observation may be technically valid while insufficient for the governance conclusion being drawn.

Evidence Provenance

Material deployment-state evidence must remain attributable.

Relevant elements include:

Evidence ID

Source

Collection Method

Collection Time

Observed Environment

Observed Segment

Integrity Reference

Processing Reference

Verification Result

Known Limitations

Evidence provenance supports reconstruction of deployment-state conclusions.

Evidence Sufficiency

DSMM must determine whether available evidence is sufficient to support the claimed deployment state.

Relevant dimensions include:

Execution Evidence

Production Version Evidence

Configuration Evidence

Operational Condition Evidence

Monitoring Coverage

Observation Freshness

Source Reliability

Evidence Consistency

Known Gaps

Required Verification Criteria

A deployment cannot be classified as operationally verified solely because

available observations contain no reported failures.

Execution–Reality Alignment

DSMM must compare execution reports with observed operational conditions.

Relevant relationships include:

Reported Installation ↔ Observed Artifact Presence

Reported Activation ↔ Observed Runtime State

Reported Version ↔ Verified Production Version

Reported Completion ↔ Observed Deployment Progression

Reported Success ↔ Operational Acceptance Conditions

Material differences must remain visible and subject to assessment.

Intended–Observed State Alignment

DSMM must evaluate whether the observed deployment state corresponds to the intended operational state.

Relevant dimensions include:

Release Identity

Artifact Identity

Environment

Activation

Configuration

Dependencies

Service Conditions

Operational Restrictions

Verification Criteria

Residual Exceptions

Alignment must be evaluated within the relevant deployment scope.

Authorization–Observed State Alignment

DSMM must identify material differences between authorized conditions and observed deployment reality.

Examples include:

Unauthorized Environment

Unexpected Deployment Scope

Unapproved Software Version

Prohibited Configuration Change

Execution Outside Authorized Window

Unauthorized Tool Action

Violation of Operational Restriction

DSMM records the observed deviation and refers authorization questions to

DAM.

Operational Condition Verification

Operational condition verification evaluates whether required conditions

are supported by available evidence.

Relevant conditions may include:

Service Availability

Functional Behavior

Performance Limits

Dependency Connectivity

Data Integrity

Security Restrictions

Operational Safety Conditions

Required Configuration

Production Version Consistency

Monitoring Availability

Verification must distinguish confirmed satisfaction from unknown or insufficiently observed conditions.

Verification Condition States

DSMM applies the following condition model:

Satisfied

Not Satisfied

Partially Satisfied

Unknown

Not Applicable

Conflicted

Exception

Each condition must preserve its relevant evidence or legitimate exception reference.

Unknown ≠ Satisfied.

No Failure Detected ≠ Condition Verified.

Deployment State Consistency

Deployment-state consistency concerns whether observed conditions

collectively support the intended and authorized deployment state.

Relevant dimensions include:

Execution Consistency

Version Consistency

Environment Consistency

Configuration Consistency

Dependency Consistency

Operational Condition Consistency

Authorization Condition Consistency

Verification Evidence Consistency

A deployment may satisfy some conditions while remaining inconsistent overall.

Partial Deployment State

A partial deployment state exists when deployment actions or resulting conditions have been established for only part of the intended scope.

Relevant elements include:

Completed Segments

Pending Segments

Failed Segments

Unknown Segments

Observed Version Distribution

Operational Impact

Applicable Continuation Conditions

Required Intervention

A partial deployment may be acceptable during an authorized staged rollout.

It must not automatically be treated as a completed deployment.

Mixed-Version Operational State

Mixed-version environments require assessment of both version distribution

and operational conditions.

Relevant considerations include:

Authorized Mixed-Version Scope

Observed Version Distribution

Compatibility Requirements

Service Continuity

Dependency Interactions

Configuration Compatibility

Operational Health

Transition Duration

Verification Coverage

PVCM establishes the version distribution.

DSMM evaluates whether the observed mixed-version state remains

operationally acceptable.

Deployment Anomaly

A deployment anomaly is an observed condition that materially differs from an expected or required deployment condition.

Examples include:

Unexpected Software Version

Activation Failure

Service Degradation

Configuration Drift

Dependency Failure

Unexpected Resource Consumption

Unverified Operational Segment

Security-Relevant Event

Unplanned Traffic Transition

Physical-System Behavior Deviation

An anomaly must be identified without prematurely assigning its cause.

Anomaly Classification

DSMM may classify anomalies according to:

Affected Condition

Operational Scope

Severity

Evidence Confidence

Persistence

Reversibility

Potential Consequence

Required Response

Current State

Anomaly classification must remain connected to observed evidence and applicable deployment conditions.

Anomaly Attribution Boundary

DSMM governs observation and deployment-state assessment.

It does not automatically establish why an anomaly occurred or who is responsible.

Relevant distinctions include:

Observed Failure ≠ Verified Root Cause.

Temporal Association ≠ Causal Attribution.

Deployment Correlation ≠ Deployment Causation.

Where root-cause analysis or accountability determination is required, DSMM preserves the relevant evidence and refers the matter to applicable mechanisms.

Monitoring Uncertainty

DSMM must preserve uncertainty affecting deployment-state conclusions.

Examples include:

Incomplete Telemetry

Unknown Production Version

Unverified Service Condition

Unobserved Device

Stale Evidence

Conflicting Metrics

Unavailable Dependency Status

Uncertain Recovery Condition

Uncertainty must influence the deployment-state classification where material.

Conflicting Observations

Monitoring sources may report inconsistent states.

Examples include:

Pipeline Reports Success

Application Health Check Reports Failure

Runtime Inventory Reports Unexpected Version

Service Metrics Report Normal Availability

DSMM must preserve the conflict and identify the affected governance conclusion.

Conflicting evidence must not be silently resolved by selecting the most favorable observation.

Monitoring Failure

Monitoring failure occurs when required observation capability becomes unavailable, unreliable, or insufficient.

Examples include:

Telemetry Loss

Monitoring Agent Failure

Logging Failure

Metric Collection Failure

Evidence Processing Failure

Observation Delay

Source Integrity Failure

Monitoring Coverage Loss

Monitoring failure may itself require intervention when it prevents verification of mandatory deployment conditions.

Monitoring Continuity

DSMM must preserve required monitoring continuity during material deployment transitions.

Relevant elements include:

Monitoring Start Condition

Required Observation Period

Monitoring Availability

Observation Frequency

Evidence Retention

Monitoring Handoff

Monitoring Failure Response

Monitoring Termination Condition

Monitoring must not be terminated solely because a deployment pipeline reports completion.

Deployment Stabilization

Deployment stabilization concerns whether the resulting operational state

remains acceptable over an applicable observation period.

Relevant elements include:

Stabilization Criteria

Required Observation Period

Operational Metrics

Version Consistency

Error Conditions

Service Availability

Material Anomalies

Monitoring Coverage

Stabilization Evidence

The required stabilization period depends on the deployment's operational

characteristics and consequences.

Operational Acceptance Assessment

DSMM must evaluate whether the deployment satisfies applicable operational acceptance criteria.

Relevant elements include:

Authorized Deployment Reference

Verified Production Version

Required Operational Conditions

Verification Results

Monitoring Coverage

Evidence Sufficiency

Material Exceptions

Residual Restrictions

Acceptance Classification

A deployment may be technically complete while operational acceptance remains pending or unsupported.

Intervention Thresholds

Intervention thresholds define conditions requiring reassessment or

corrective action.

Examples include:

Critical Service Failure

Unacceptable Error Rate

Production Version Mismatch

Security Restriction Violation

Data Integrity Failure

Monitoring Coverage Below Required Level

Unauthorized Deployment Action

Unacceptable Physical-System Behavior

Recovery Threshold Exceeded

Thresholds must remain associated with the relevant deployment and operational conditions.

Intervention Trigger Evaluation

DSMM evaluates whether observed conditions satisfy an intervention trigger.

Relevant elements include:

Trigger ID

Observed Condition

Threshold

Evidence

Observation Time

Affected Scope

Required Response

Decision Authority Reference

Escalation Requirement

A detected trigger must not automatically be treated as authorization for every possible corrective action.

Deployment Continuation Assessment

DSMM may support continuation decisions where deployment occurs in stages.

Relevant considerations include:

Current Deployment State

Required Checkpoint Conditions

Operational Health

Version Consistency

Monitoring Coverage

Material Anomalies

Applicable Authorization

Remaining Execution Scope

Required Restrictions

DSMM provides evidence and classification.

The applicable decision authority determines whether continuation is permitted.

Deployment Restriction and Pause

A deployment may require restriction or pause when conditions cannot be sufficiently established or maintained.

Relevant triggers include:

Material Monitoring Gap

Unknown Production Version

Critical Operational Anomaly

Authorization Condition Deviation

Unresolved Evidence Conflict

Unacceptable Service Degradation

Recovery Readiness Concern

DSMM identifies the relevant state and required intervention reference.

It does not independently grant authority to perform the intervention.

Rollback Trigger Reference

DSMM must identify conditions potentially requiring rollback consideration.

Examples include:

Failed Operational Acceptance

Unacceptable Version Inconsistency

Critical Service Degradation

Deployment-Induced Configuration Failure

Unacceptable Dependency Failure

Material Security Condition

Failed Stabilization

The rollback decision and execution remain governed by RRVM and applicable

authority mechanisms.

Recovery Trigger Reference

Recovery may be required when rollback is unavailable, insufficient, or inappropriate.

Relevant conditions include:

Irreversible Data Change

Infrastructure Failure

Unrecoverable Previous Version

Dependency Incompatibility

Partial System Restoration

Persistent Operational Degradation

Physical-System Recovery Requirement

DSMM preserves the observed conditions supporting recovery assessment.

Autonomous Monitoring Systems

Autonomous agents may participate in deployment monitoring.

Relevant governance elements include:

Agent Identity

Assigned Monitoring Function

Observation Sources

Monitoring Scope

Permitted Analysis

Decision Boundaries

Evidence Requirements

Escalation Conditions

Human Override Reference

An agent-generated monitoring conclusion must remain distinguishable from independently verified operational evidence.

Autonomous Intervention Boundary

An autonomous monitoring agent may detect a condition requiring intervention.

Detection does not automatically establish authority to execute rollback, modify infrastructure, disable services, or change deployment scope.

Monitoring Authority ≠ Intervention Authority.

Anomaly Detection ≠ Unrestricted Corrective Permission.

Any autonomous intervention must remain within the applicable authority, permission, execution, and operational boundaries.

Material-Change Reassessment

Deployment-state reassessment may be required when:

Production Version Changes

Configuration Changes

Dependencies Change

Operational Environment Changes

Monitoring Coverage Changes

Authorization Conditions Change

New Anomalies Emerge

Recovery Actions Occur

Observation Evidence Becomes Stale

Operational Restrictions Change

Reassessment must identify whether previous deployment-state conclusions remain supported.

Historical Deployment State Continuity

DSMM must preserve relevant historical deployment-state classifications.

These may include:

Execution In Progress

Partially Established

Verification Pending

Operationally Established

Degraded

Intervention Required

Rollback Initiated

Recovery In Progress

Superseded

Unknown

Historical states must remain associated with their observation times and evidence.

A later acceptable state must not erase earlier material deviations.

Deployment State Governance Classification

DSMM recognizes the following principal classifications:

Not Observed

Monitoring Pending

Execution In Progress

Partially Established

Verification Pending

Verification In Progress

Operationally Established

Operationally Established With Restrictions

Degraded

Operationally Unacceptable

Monitoring Insufficient

Evidence Insufficient

State Inconsistent

State Conflicted

Intervention Required

Deployment Paused

Rollback Assessment Required

Recovery Assessment Required

Reassessment Required

Superseded

Unknown

Each classification must remain associated with the relevant deployment, environment, scope, and evidence time.

Governance Requirements

A conforming DSMM implementation must satisfy the following requirements. Identity and References

• Establish deployment-state identity

• Identify the relevant deployment

• Preserve the deployment plan reference

• Preserve applicable authorization conditions

• Identify the target environment

• Preserve production version references

• Define the monitoring scope

• Identify required operational conditions

Observation and Evidence

• Identify material observation sources

• Preserve observation times

• Preserve observation provenance

• Evaluate source reliability

• Evaluate monitoring coverage

• Identify monitoring blind spots

• Evaluate evidence freshness

• Evaluate evidence sufficiency

• Preserve conflicting observations

• Preserve material uncertainty

State Assessment

• Observe deployment progression

• Distinguish execution completion from operational establishment

• Evaluate intended–observed state alignment

• Evaluate authorization–observed state alignment

• Evaluate production version consistency references

• Evaluate applicable operational conditions

• Preserve partial deployment states

• Evaluate mixed-version operational conditions

• Identify material anomalies

• Avoid unsupported causal attribution

• Evaluate stabilization requirements

• Establish operational acceptance classification

Intervention and Continuity

• Identify intervention thresholds

• Evaluate intervention triggers

• Preserve continuation assessment references

• Identify restriction or pause conditions

• Preserve rollback trigger references

• Preserve recovery trigger references

• Maintain monitoring continuity

• Identify material reassessment triggers

• Preserve historical deployment states

• Provide evidence-supported handoff to RRVM

Governance Logic

Deployment Identity → Intended State → Authorization Conditions →

Production Version Reference → Monitoring Scope → Observation Sources → Execution Observations → Operational Observations → Coverage Assessment → Evidence Sufficiency → Intended–Observed Comparison → Condition Verification → Anomaly Assessment → Deployment State Classification → Continue / Restrict / Pause / Escalate / Rollback Assessment / Recovery Assessment

Where monitoring is insufficient:

Required Condition → Missing or Unreliable Observation → Evidence Gap →

Monitoring Insufficient / Evidence Insufficient → Reassessment or Intervention Reference

Where material anomalies occur:

Observed Deviation → Evidence Assessment → Operational Consequence →

Threshold Evaluation → Intervention Reference → State Reclassification

Where conditions change:

Previously Supported State → Material Change → New Observation → Evidence

Reassessment → Updated State Classification

DSMM must not convert technical completion reports into operational acceptance without sufficient evidence.

Governance Outputs

The principal output is:

Software Deployment State Profile

Supporting outputs may include:

Deployment State Identity

Monitoring Scope Record

Observation Source Register

Deployment Progression Record

Production Version Reference

Operational Condition Assessment

Monitoring Coverage Assessment

Monitoring Blind Spot Register

Evidence Freshness Assessment

Evidence Sufficiency Assessment

Intended–Observed State Comparison

Authorization–Observed State Comparison

Deployment Anomaly Register

Monitoring Conflict Record

Deployment Stabilization Assessment

Operational Acceptance Classification

Intervention Trigger Record

Deployment Restriction Reference

Rollback Assessment Trigger

Recovery Assessment Trigger

Historical Deployment State Register

Deployment State Reassessment Record

RRVM Handoff Record

Minimum Implementation Framework

A minimum DSMM implementation must establish eight governance layers. Layer 1 — Deployment State Identity

Identify the deployment, environment, production version references, and

monitored operational scope. Layer 2 — Intended and Authorized Conditions

Preserve the intended deployment state, operational acceptance criteria,

and applicable authorization restrictions. Layer 3 — Observation

Collect relevant execution, version, configuration, dependency, and

operational condition evidence. Layer 4 — Monitoring Coverage

Identify which required conditions and operational segments are observable and where material gaps remain. Layer 5 — Verification

Compare observed conditions against intended and authorized states using

evidence of sufficient relevance and reliability. Layer 6 — Anomaly and Intervention

Identify material deviations, monitoring failures, intervention thresholds,

and escalation requirements. Layer 7 — Classification and Continuity

Establish the deployment-state classification, preserve historical

observations, and reassess material changes. Layer 8 — Governed Output and Handoff

Establish the Software Deployment State Profile and provide relevant

intervention, rollback, or recovery references to RRVM.

The implementation may use manual observation, monitoring platforms, automated analysis, autonomous agents, or other suitable mechanisms.

The methodological requirements remain independent of the chosen monitoring

technology.

Cross-Module Boundary

DPM → DSMM

DPM establishes intended deployment conditions and verification

requirements.

DSMM evaluates whether the observed deployment state satisfies those

conditions.

DAM → DSMM

DAM establishes authorization conditions and restrictions.

DSMM identifies observed deviations from those conditions and supports

authorization reassessment where required.

PVCM → DSMM

PVCM establishes production version identity and consistency.

DSMM evaluates the operational consequences of the observed production

version state.

DSMM → RRVM

DSMM identifies unacceptable conditions, intervention triggers, and

evidence supporting rollback or recovery assessment.

RRVM governs corrective action selection, authority, execution, and

recovery verification.

Deployment State Monitoring ≠ Rollback Execution.

Cross-Architecture Boundary

DSMM consumes relevant governance references from:

SQS — Software Quality Standard

Quality conditions, verification criteria, and defect references.

SSS — Software Security Standard

Security restrictions, vulnerability conditions, and remediation references.

SDS — Software Dependency Standard

Dependency identity and operational dependency conditions.

SCFS — Software Configuration Standard

Configuration identity and consistency references.

SCMS — Software Change Management Standard

Change coordination and lifecycle re-entry references.

SLCS — Software Lifecycle Control Standard

Lifecycle state consistency, transitions, exceptions, and continuity.

ASGA™ — Autonomous Systems Governance Architecture

Autonomous observation, intervention, and override boundaries.

DSMM does not replace these governance mechanisms.

It evaluates their deployment-state relevance where applicable.

Practical Applications

Continuous Delivery

Verifying that automated deployment completion corresponds to an acceptable operational state.

Cloud-Native Systems

Monitoring deployment progression, service health, version distribution,

and operational conditions across clusters and regions.

Autonomous AI Deployment

Preventing autonomous systems from treating self-reported success or

incomplete telemetry as sufficient operational verification.

Critical Infrastructure

Establishing monitoring coverage, intervention thresholds, and

evidence-supported acceptance of consequential software updates.

Robotics

Monitoring software deployment states where updates may affect physical movement, equipment behavior, or operational safety.

Distributed Device Fleets

Identifying unobserved devices, partial rollouts, unknown versions, and

inconsistent deployment outcomes.

Use Case 1 — Deployment Pipeline Reports Success While Production

Service Is Degraded

A software provider deploys an authorized release into production.

The deployment pipeline reports successful completion.

The expected artifact is present and activated.

However, application response times increase substantially, and a critical

service dependency begins failing.

DSMM evaluates:

Execution Completion → Production Version → Operational Metrics → Required

Conditions → Monitoring Coverage → Operational Acceptance

The assessment identifies:

• Deployment execution completed

• The intended software version is present

• Critical operational conditions are not satisfied

• The deployment cannot be classified as operationally established

• Intervention assessment is required

The appropriate classification is:

Operationally Unacceptable — Intervention Required.

Deployment Executed ≠ Deployment Successfully Established.

Use Case 2 — A Robotic Fleet Reports Healthy Devices but Monitoring

Coverage Is Incomplete

A robotics operator deploys a software update across autonomous mobile

robots.

The management system reports healthy operation for connected devices.

Several robots are temporarily disconnected and provide no current telemetry.

The platform displays an overall successful deployment state.

DSMM evaluates:

Deployment Scope → Device Coverage → Observed Conditions → Missing

Telemetry → Evidence Sufficiency → Deployment State Classification

The assessment identifies:

• Connected devices provide acceptable operational observations

• Disconnected devices remain unobserved

• The monitoring scope does not cover the complete fleet

• The available evidence cannot support a fleet-wide operational

acceptance claim.

The appropriate classification is:

Monitoring Insufficient — Partial Operational Verification.

Healthy Observed Devices ≠ Entire Fleet Operationally Verified.

Use Case 3 — An Autonomous Agent Interprets a Monitoring Alert as

Permission to Roll Back

An autonomous deployment agent monitors a production rollout.

The agent detects an elevated application error rate.

Its assigned function permits deployment monitoring and escalation.

The agent attempts to execute a full production rollback using available

infrastructure credentials.

DSMM evaluates:

Observed Anomaly → Evidence → Intervention Threshold → Agent Monitoring

Scope → Required Authority → Escalation

The assessment identifies:

• The anomaly is supported by monitoring evidence

• The intervention threshold has been reached

• The agent is authorized to monitor and escalate

• The agent's monitoring authority does not independently authorize

• RRVM and applicable authority mechanisms must govern the corrective

rollback execution. action.

The appropriate classification is:

Intervention Required — Rollback Authorization Assessment.

Monitoring Authority ≠ Intervention Authority.

Critical Distinctions

Deployment Executed ≠ Deployment Successfully Established.

Monitoring Active ≠ Deployment State Fully Observable.

Service Available ≠ Deployment Operationally Acceptable.

No Alert ≠ No Material Problem.

Observed Condition ≠ Verified Cause.

Monitoring Configured ≠ Monitoring Sufficient.

Unknown ≠ Satisfied.

No Failure Detected ≠ Condition Verified.

Observed Failure ≠ Verified Root Cause.

Temporal Association ≠ Causal Attribution.

Deployment Correlation ≠ Deployment Causation.

Pipeline Completion ≠ Operational Acceptance.

Production Version Verified ≠ Deployment State Fully Verified.

Partial Deployment ≠ Completed Deployment.

Monitoring Authority ≠ Intervention Authority.

Anomaly Detection ≠ Unrestricted Corrective Permission.

Deployment State Monitoring ≠ Rollback Execution.

Healthy Observed Devices ≠ Entire Fleet Operationally Verified.

Canonical Boundary

DSMM governs the observation, evidence assessment, verification,

classification, and reassessment of actual software deployment states within defined operational environments.

It establishes monitoring scope, observation requirements, evidence

coverage, operational condition verification, intended–observed state alignment, anomaly identification, intervention trigger references, and historical deployment-state continuity.

DSMM does not authorize deployments, independently establish production artifact identity, execute corrective actions, or determine recovery success outside applicable governance mechanisms.

Its principal governance output is the:

Software Deployment State Profile

This output provides the evidence-supported deployment-state basis for

operational intervention, rollback or recovery assessment, and relevant software lifecycle governance.

Canonical Closing Statement

A deployment is not successfully established merely because its execution has finished or its monitoring dashboard appears healthy.

A pipeline may report success while the deployed service is degraded.

A software version may be correct while its configuration is incompatible with the operational environment.

A monitoring system may report normal conditions while critical devices remain unobserved.

An autonomous agent may detect a deployment failure without possessing authority to execute corrective actions.

And a previously verified deployment state may become unacceptable when operational conditions materially change.

DSMM establishes the governance structure required to distinguish these

realities.

It requires deployment-state conclusions to be supported by observations

that are relevant, sufficiently complete, reliable, and current.

It preserves monitoring blind spots, conflicting evidence, and unresolved

conditions instead of allowing them to disappear behind simplified success indicators.

It connects deployment execution, production version identity, operational

verification, and intervention thresholds without collapsing their separate governance responsibilities.

And it ensures that operational acceptance remains a verifiable conclusion rather than an assumption derived from technical execution.

The purpose of deployment monitoring is not merely to observe whether software is running. It is to establish whether the actual deployment state remains consistent with its intended, authorized, and operationally acceptable conditions—and to identify when that conclusion can no longer be supported.

DSMM transforms deployment monitoring from passive technical observation

into an evidence-supported governance mechanism for operational reality, intervention readiness, and software lifecycle continuity.